

RUSHIKESH MOHALKAR

AI / ML ENGINEER

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SUMMARY

AI/ML Engineer focused on **Generative AI, Deep Learning, Reinforcement Learning, and production AI systems**. Experienced in building **RAG pipelines, LLM applications, AI agents, QLoRA fine-tuning, inference optimization, and self-hosted AI systems** using Python, PyTorch, Hugging Face, LangChain, LangGraph, Docker, FastAPI, and AWS.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash
ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, Transformers
GenAI/LLMs: RAG, LangChain, LangGraph, LlamaIndex, Prompt Engineering, QLoRA, Multi-Agent Systems, Ollama
NLP: NLTK, NER, Text Classification, Sequence Modeling
MLOps/Cloud: AWS (S3, EC2, SageMaker), Docker, Kubernetes, FastAPI, Flask, Nginx, Git, Linux
Core: Reinforcement Learning (PPO), Transfer Learning, LLM Optimization, Latency & Cost Optimization, Time-Series Forecasting

PROJECTS

- [Enterprise RAG & AI Operations Platform](#) — *LangGraph, FastAPI, Qdrant, PostgreSQL, Redis, BM25, Docker*
- Built a production-oriented RAG system combining **dense + BM25 hybrid retrieval, reranking, HyDE, CRAG, and Self-RAG** with corrective query rewriting and grounded response generation.
 - Implemented **Text2SQL with human approval, semantic caching, nine-layer input/output guardrails, evaluation pipelines, and REST APIs**, with retrieval evaluation achieving **0.76 precision@5** on the project benchmark.
- [Custodian — Multi-Agent Accounts Payable AI](#) — *Python, FastAPI, LiteLLM, Docker, React, MLflow*
- Built a governed multi-agent system for **invoice ingestion, fraud/risk scoring, approval routing, and automated payment**, integrating real LLM providers through LiteLLM.
 - Implemented **PII redaction, policy enforcement, audit logging, OCR ingestion, role-based API authentication, MLflow tracking, Prometheus metrics, and React dashboards** for auditable AI operations.
- [RAGApp](#) — *LangChain, FAISS, FLAN-T5, Hugging Face, Gradio*
- Built an end-to-end **RAG pipeline** for PDF ingestion, semantic chunking, embeddings, FAISS retrieval, and context-grounded question answering.
 - Optimized retrieval and prompting to reduce **inference latency and token usage**, with a local/open-source inference workflow and interactive Gradio interface.
- [Oryza — Self-Hosted LLM Chatbot](#) — *Ollama, Qwen 2.5, Flask, Docker*
- Built a private offline chatbot serving **Qwen 2.5 1.5B** locally through Ollama and Flask, eliminating dependency on external LLM APIs.
 - Containerized the application using **Docker Compose and Nginx** and optimized local inference for resource-constrained deployment.

PROFESSIONAL EXPERIENCE

- Programmer Analyst — Cognizant** Jul 2024 – Present
- Built Python-based **data and defect-analysis automation pipelines**, reducing manual execution time by **15%** and accelerating delivery cycles.
 - Partnered with engineering teams throughout the release lifecycle, improving validation and reporting and reducing release delays by **20%**.

EDUCATION

B.E. — Electronics & Telecommunications Engineering AISSMS IOIT, Savitribai Phule Pune University, Pune

CERTIFICATIONS

Mathematical Foundations of Machine Learning | Python for Data Science & ML Bootcamp | Deep Learning A-Z | AI Engineer Core Track — LLMs, RAG, QLoRA & Agents | Generative AI with LangChain & Hugging Face